Unit 02.A Handout: Visualization

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Example 1: Histogram

The mile-run times of 50 students were recorded in minutes. Use the frequency table below to complete the relative-frequency column and draw a histogram.

4:55.21 Jake, 5:28.20 Alyssa, 7:15.03 Hannah, ..., 9:31.07 Tyler.

The class frequencies are given below:

Class	**Frequency**	**Relative Frequency**
$[4, 5)$	2	
$[5, 6)$	5	
$[6, 7)$	7	
$[7, 8)$	11	
$[8, 9)$	14	
$[9, 10)$	11	
Total	50	

Example 2: Dot Plot

The following data are breaking strengths, measured in gigapascals (GPa), for $n = 15$ carbon fibers. Construct a dot plot.

$$\{0.1,\ 2.0,\ 2.3,\ 2.5,\ 3.2,\ 3.4,\ 3.4,\ 3.7,\ 3.9,\ 4.0,\ 4.1,\ 4.2,\ 4.2,\ 4.3,\ 4.4\}$$

Example 3: Stem-and-Leaf Plot

(a)

Using the same data (Example 2), draw the stem-and-leaf plot.

$$\{0.1,\ 2.0,\ 2.3,\ 2.5,\ 3.2,\ 3.4,\ 3.4,\ 3.7,\ 3.9,\ 4.0,\ 4.1,\ 4.2,\ 4.2,\ 4.3,\ 4.4\}$$

(b)

Suppose, we have collected a random sample of 15 observations, 7 were from Method A and 8 were from Method B. Considering the following data, draw a side-by-side Stem-and-Leaf Plot. Describe the plots as right-skewed, left-skewed, or symmetric. Also, compare the mean and median of the observations of Method A and Method B.

Method A: $n = 7$	**Method B:** $n = 8$
$\{0.1,\ 2.0,\ 2.5,\ 3.2,\ 3.4,\ 3.4,\ 4.0\}$	$\{2.3,\ 3.7,\ 3.9,\ 4.1,\ 4.2,\ 4.2,\ 4.3,\ 4.4\}$

Example 4: Box Plot

Suppose, we have the following $n = 7$ observations:

$$\{0,\ 11,\ 13,\ 25,\ 34,\ 40,\ 81\}$$

For the above data set, find the values of the lower quartile, median, upper quartile, IQR, step, UIF, LIF, UOF, and LOF. Use this information to draw a boxplot.